

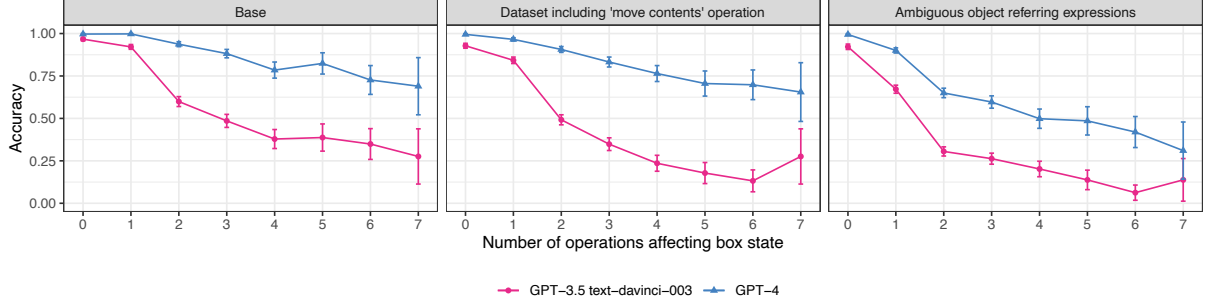


Figure 8: Entity tracking accuracy of text-davinci-003 and GPT-4 in 2-shot in-context learning experiments.

C Implementation Details

In-context For GPT-3, we used the OpenAI API. We used greedy decoding with the temperature parameter set to 0, used a maximum target generation length of 150, and used the newline character as an additional stop token. Inference time and model size is not available for GPT-3, but we generated about 16 million tokens in total. All GPT-3 experiments that we report here in the paper were conducted in January 2023.

For Flan-T5, we used a beam size of 3 and a maximum target generation length of 256. Other hyperparameters were kept as the default values of T5ConditionalGeneration in the HuggingFace library. Inference for T5-XL took about 5 hours on a single A100 GPU, and for T5 base, about 2 hours.

Finetuning We finetuned T5 for a single epoch using a batch size of 8 and a learning rate of 1×10^{-4} . In initial explorations, increasing the number of finetuning epochs did not yield substantial gains on development set performance, and was sometimes even harmful. Training and inference took around 3 hours on a single RTX8000 GPU.

D Additional Results

D.1 GPT-3.5 Zero-shot Results

As mentioned in the main text, GPT-3.5 was able to output the contents of boxes in the correct format without any in-context demonstrations (see Table 5 for the prompt template).

Figure 9 compares the performance of text-davinci-003 in the 2-shot setting to the zero-shot setting. Model performance degraded slightly without demonstration examples, which suggests that the examples are indeed helpful in guiding the model to correctly perform the task. Nevertheless, we observed non-negligible performance even in the zero-shot setting, which

corroborates the conclusion that GPT 3.5 exhibits non-trivial entity tracking capacities.

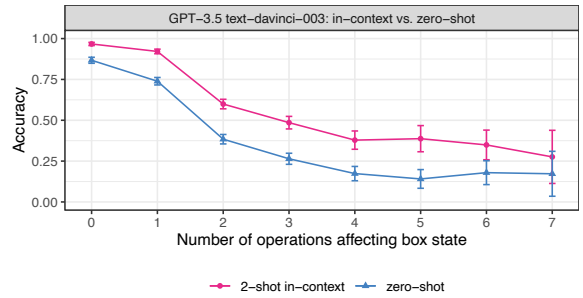


Figure 9: Entity tracking accuracy of GPT 3.5 under in-context (2-shot) and zero-shot settings.

D.2 GPT-4 Results

We also probed the more recently released GPT-4 model (OpenAI, 2023) through the OpenAI API. Figure 8 compares the performance of GPT-3.5 text-davinci-003 and GPT-4 on the base dataset (left) as well as the more challenging MoveContents (middle) and AmbiRef (right) datasets discussed in Section 4.2. Across all datasets, GPT-4 performed considerably better than GPT-3.5. But similarly to GPT-3.5, performance degraded as more operations affected a box. Considering that no public information about the training data, training procedure or architectural details about GPT-4 exists, we cannot draw any conclusions about the reason behind the improved performance of GPT-4 compared to previous models.

E Prompts

Table 3 shows the 2-shot prompts used for in-context experiments. Table 4 shows the 2-shot prompts used for the AltForms experiments, where the demonstrations contain descriptions that have lower lexical overlap with the test examples. Table 5 shows the zero-shot prompts discussed in Appendix D. All prompts are available at <https://>

github.com/sebschu/entity-tracking-lms.

F Dataset Statistics and License Information

See Tables 6 and 7 for descriptive statistics of our datasets. All datasets are released under the GNU General Public License v3.0.

2-shot prompt with all boxes queried at once (GPT-3 experiments)

Given the description after "Description:", write a true statement about all boxes and their contents to the description after "Statement:".

Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map.

Statement: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map.

Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map. Remove the car from Box 0. Remove the paper and the string from Box 3. Put the plane into Box 0. Move the map from Box 6 to Box 2. Remove the bill from Box 4. Put the coat into Box 3.

Statement: Box 0 contains the plane, Box 1 contains the cross, Box 2 contains the bag and the machine and the map, Box 3 contains the coat, Box 4 contains nothing, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle.

Description: {description}

Statement: Box 0 contains

2-shot prompt with boxes queried individually (T5 experiments)

Given the description after "Description:", write a true statement about a box and the contents of this box according to the description after "Statement:".

Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map.

Statement: Box 1 contains the cross.

Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map. Remove the car from Box 0. Remove the paper and the string from Box 3. Put the plane into Box 0. Move the map from Box 6 to Box 2. Remove the bill from Box 4. Put the coat into Box 3.

Statement: Box 2 contains the bag and the machine and the map.

Description: {description}

Statement: Box {boxnum} contains

Table 3: Prompts with 2-shot in-context demonstrations.